



ENDOCRINOLOGY

Adrenal Gland - 2

Internal Medicine Lecture Series

Adrenal insufficiency

- ◆ Adrenal insufficiency results from inadequate secretion of cortisol and/or aldosterone.
- ◆ It is potentially fatal and variable in its presentation.
- ◆ The most common is ACTH deficiency (secondary adrenocortical failure), usually because of inappropriate withdrawal of chronic glucocorticoid therapy or a pituitary tumour.
- ◆ Congenital adrenal hyperplasias and Addison's disease (primary adrenocortical failure) are rare causes.

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20.40 Causes of adrenocortical insufficiency

Secondary (↓ACTH)

- Withdrawal of suppressive glucocorticoid therapy
- Hypothalamic or pituitary disease

Primary (↑ACTH)

Addison's disease

Common causes

- Autoimmune: Sporadic
- Polyendocrine syndromes (APS types 1 and 2)
- Tuberculosis
- HIV/AIDS
- Metastatic carcinoma
- Bilateral adrenalectomy

Rare causes

- Lymphoma
- Intra-adrenal haemorrhage (Waterhouse–Friderichsen syndrome following meningococcal sepsis)
- Amyloidosis
- Haemochromatosis

Corticosteroid biosynthetic enzyme defects

- Congenital adrenal hyperplasias
- Drugs: metyrapone, ketoconazole, etomidate

(ACTH = adrenocorticotrophic hormone; APS = autoendocrine polyendocrine syndromes; HIV/AIDS = human immunodeficiency virus/acquired immune deficiency syndrome)

- ◆ In Addison's disease, either glucocorticoid or mineralocorticoid deficiency may come first, but eventually all patients fail to secrete both classes of corticosteroid.
- ◆ Patients may present with chronic features and/or in acute circulatory shock.
- ◆ With a chronic presentation, initial symptoms are often misdiagnosed as chronic fatigue syndrome or depression.
- ◆ Adrenocortical insufficiency should also be considered in patients with hyponatraemia, even in the absence of symptoms.
- ◆ Features of an acute adrenal crisis include circulatory shock with severe hypotension, hyponatraemia, hyperkalaemia and, in some instances, hypoglycaemia and hypercalcaemia. Muscle cramps, nausea, vomiting, diarrhoea and unexplained fever may be present.
- ◆ The crisis is often precipitated by intercurrent disease, surgery or infection.
- ◆ Vitiligo occurs in 10–20% of patients with autoimmune Addison's disease

Investigations

- ◆ Treatment should not be delayed to wait for results in patients with suspected acute adrenal crisis.
- ◆ Here, a random blood sample should be stored for subsequent measurement of cortisol and, if the patient's clinical condition permits, it may be appropriate to spend 30 minutes performing a short ACTH stimulation test before administering hydrocortisone.

1- Assessment of glucocorticoids:

- ◆ Random plasma cortisol is usually low in patients with adrenal insufficiency but it may be within the reference range, yet inappropriately low, for a seriously ill patient.
- ◆ Random measurement of plasma cortisol cannot therefore be used to confirm or refute the diagnosis, unless the value is above 500 nmol/L ($> 18 \mu\text{g/dL}$), which effectively excludes adrenal insufficiency.
- ◆ More useful is the short ACTH stimulation test (also called the tetracosactrin or short Synacthen test).



secondary adrenal insufficiency. These can be distinguished by measurement of ACTH (which is low in ACTH deficiency and high in Addison's disease).

2- Assessment of mineralocorticoids

- ◆ Mineralocorticoid secretion in patients with suspected Addison's disease cannot be adequately assessed by electrolyte measurements since hyponatraemia occurs in both aldosterone and cortisol deficiency.
- ◆ Hyperkalaemia is common, but not universal, in aldosterone deficiency. Plasma renin and aldosterone should be measured in the supine position. In mineralocorticoid deficiency, plasma renin activity is high, with plasma aldosterone being either low or in the lower part of the reference range.

3- Assessment of adrenal androgens

- ◆ This is not necessary in men because testosterone from the testes is the principal androgen. In women, dehydroepiandrosterone (DHEA) and androstenedione may be measured in a random specimen of blood, though levels are highest in the morning.

4- Other tests to establish the cause

- ◆ Patients with unexplained secondary adrenocortical insufficiency should be investigated as In patients with elevated ACTH, further tests are required to establish the cause of Addison's disease.
- ◆ Adrenal autoantibodies are frequently positive in autoimmune adrenal failure. If antibody tests are negative, imaging of the adrenal glands with CT or MRI is indicated.
- ◆ Tuberculosis causes adrenal calcification visible on plain X-ray or ultrasound scan. An HIV test should be performed if risk factors for infection are present .
- ◆ Adrenal metastases are a rare cause of adrenal insufficiency.
- ◆ Patients with evidence of autoimmune adrenal failure should be screened for other organ-specific autoimmune diseases, such as thyroid disease, pernicious ;

- ◆ Patients with adrenocortical insufficiency always need glucocorticoid replacement therapy and usually, but not always, mineralocorticoid therapy.
- ◆ There is some evidence that adrenal androgen replacement may also be beneficial in women. Other treatments depend on the underlying cause.

1- Glucocorticoid replacement

- ◆ Adrenal replacement therapy consists of oral hydrocortisone (cortisol) 15– 20 mg daily in divided doses, typically 10 mg on waking and 5 mg at around 1500 hrs.
- ◆ These are physiological replacement doses which should not cause Cushingoid side-effects.
- ◆ The dose may need to be adjusted for the individual patient but this is subjective. Excess weight gain usually indicates overreplacement, whilst persistent lethargy or hyperpigmentation may be due to an inadequate dose or lack of absorption.
- ◆ Measurement of plasma cortisol levels is not usually helpful.

2- Mineralocorticoid replacement

- ◆ Fludrocortisone (9a-fluoro-hydrocortisone) is administered at the usual dose of 0.05– 0.15 mg daily, and adequacy of replacement may be assessed by measurement of blood pressure, plasma electrolytes and plasma renin.

3- Androgen replacement

- ◆ Androgen replacement with DHEA (50 mg/day) is occasionally given to women with primary adrenal insufficiency who have symptoms of reduced libido and fatigue, but the evidence in support of this is not robust and treatment may be associated with side-effects such as acne and hirsutism.

20.43 Management of adrenal crisis

Correct volume depletion

- IV saline as required to normalise blood pressure and pulse
- In severe hyponatraemia (<125 mmol/L) avoid increases of plasma Na >10 mmol/L/day to prevent pontine demyelination
- Fludrocortisone is not required during the acute phase of treatment

Replace glucocorticoids

- IV hydrocortisone succinate 100 mg stat, and 100 mg 4 times daily for first 12–24 hrs
- Continue parenteral hydrocortisone (50–100 mg IM 4 times daily) until patient is well enough for reliable oral therapy

Correct other metabolic abnormalities

- Acute hypoglycaemia: IV 10% glucose
- Hyperkalaemia: should respond to volume replacement but occasionally requires specific therapy (see Box 19.17)

Identify and treat underlying cause

- Consider acute precipitant, such as infection
- Consider adrenal or pituitary pathology (see Box 20.40)

(IM = intramuscular; IV = intravenous)

Incidental adrenal mass

- ◆ It is not uncommon for a mass in the adrenal gland to be identified on a CT or MRI scan of the abdomen that has been performed for another indication.
- ◆ Such lesions are known as adrenal 'incidentalomas'.
- ◆ They are present in up to 10% of adults and the prevalence increases with age.
- ◆ Eighty-five per cent of adrenal incidentalomas are non-functioning adrenal adenomas.
- ◆ The remainder includes functional tumours of the adrenal cortex (secreting cortisol, aldosterone or androgens), pheochromocytomas, primary and secondary carcinomas, hamartomas and other rare disorders, including granulomatous infiltrations.

- ◆ There are two key questions to be resolved: is the lesion secreting hormones, and is it benign or malignant?
- ◆ Patients with an adrenal incidentaloma are usually asymptomatic. However, clinical signs and symptoms of excess glucocorticoids ,mineralocorticoids catecholamines ,and, in women, androgens should be sought. Investigations should include a dexamethasone suppression test, urine or plasma metanephrines and, in virilised women, measurement of serum testosterone, DHEA and androstenedione. Patients with hypertension should be investigated for mineralocorticoid excess.
- ◆ CT and MRI are equally effective in assessing the malignant potential of an adrenal mass,

Management

- ◆ Functional lesions and tumours of more than 4 cm in diameter should be considered for removal by adrenal surgery.
- ◆ In patients with radiologically benign, nonfunctioning lesions of less than 4 cm in diameter, surgery is only required if serial imaging suggests tumour growth.
- ◆ Optimal management of patients with lowgrade cortisol secretion, as demonstrated by the dexamethasone suppression test, remains to be established.

Primary hyperaldosteronism

- ◆ Estimates of the prevalence of primary hyperaldosteronism vary according to the screening tests employed, but it may occur in as many as 10% of people with hypertension.
- ◆ Indications to test for mineralocorticoid excess in hypertensive patients include hypokalaemia (including hypokalaemia induced by thiazide diuretics), poor control of blood pressure with conventional therapy, a family history of early-onset hypertension, or presentation at a young age.

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20.45 Causes of mineralocorticoid excess

With renin high and aldosterone high (secondary hyperaldosteronism)

- Inadequate renal perfusion (diuretic therapy, cardiac failure, liver failure, nephrotic syndrome, renal artery stenosis)
- Renin-secreting renal tumour (very rare)

With renin low and aldosterone high (primary hyperaldosteronism)

- Adrenal adenoma secreting aldosterone (Conn syndrome)
- Idiopathic bilateral adrenal hyperplasia
- Glucocorticoid-suppressible hyperaldosteronism (rare)

With renin low and aldosterone low (non-aldosterone-dependent activation of mineralocorticoid pathway)

- Ectopic ACTH syndrome
- Liquorice misuse (inhibition of 11β -HSD2)
- Liddle syndrome
- 11-deoxycorticosterone-secreting adrenal tumour
- Rare forms of congenital adrenal hyperplasia and 11β -HSD2 deficiency

(11β -HSD2 = 11β -hydroxysteroid dehydrogenase type 2; ACTH = adrenocorticotrophic hormone)

Clinical features

- ◆ Individuals with primary hyperaldosteronism are usually asymptomatic but may have features of sodium retention or potassium loss.
- ◆ Sodium retention may cause oedema, while hypokalaemia may cause muscle weakness (or even paralysis, especially in Chinese), polyuria (secondary to renal tubular damage, which produces nephrogenic diabetes insipidus) and occasionally tetany (because of associated metabolic alkalosis and low ionised calcium). Blood pressure is elevated but accelerated phase hypertension is rare.

- ◆ Routine blood tests may show a hypokalaemic alkalosis.
- ◆ Sodium is usually at the upper end of the reference range in primary hyperaldosteronism, but is characteristically low in secondary hyperaldosteronism (because low plasma volume stimulates antidiuretic hormone (ADH) release and high angiotensin II levels stimulate thirst).
- ◆ The key measurements are plasma renin and aldosterone ,and in many centres, the aldosterone:renin ratio (ARR) is employed as a screening test for primary hyperaldosteronism in hypertensive patients.
- ◆ Almost all antihypertensive drugs interfere with this ratio (β -blockers inhibit whilst thiazide diuretics stimulate renin secretion).
- ◆ Thus, individuals with an elevated ARR require further testing after stopping antihypertensive drugs for at least 2 weeks.
- ◆ If necessary, antihypertensive agents that have minimal effects on the renin–angiotensin system, such as calcium antagonists and α -blockers, may be substituted. Oral potassium supplementation may also be required, as hypokalaemia itself suppresses renin activity.

Imaging and localisation

Imaging with CT or MRI

- ◆ Mineralocorticoid receptor antagonists (spironolactone and eplerenone) are valuable in treating both hypokalaemia and hypertension in all forms of mineralocorticoid excess.
- ◆ Up to 20% of males develop gynaecomastia on spironolactone. Amiloride (10–40 mg/day), which blocks the epithelial sodium channel regulated by aldosterone, is an alternative.
- ◆ In patients with an APA, medical therapy is usually given for a few weeks to normalise whole-body electrolyte balance before unilateral adrenalectomy.
- ◆ Laparoscopic surgery cures the biochemical abnormality but, depending on the pre-operative duration, hypertension remains in as many as 70% of cases, probably because of irreversible damage to the systemic microcirculation.

Phaeochromocytoma and paraganglioma

- ◆ These are rare neuro-endocrine tumours that may secrete catecholamine (adrenaline/epinephrine, noradrenaline/ norepinephrine).
- ◆ Approximately 80% of these tumours occur in the adrenal medulla (phaeochromocytomas), while 20% arise elsewhere in the body in sympathetic ganglia (paragangliomas). Most are benign but approximately 15% show malignant features.
- ◆ Around 30% are associated with inherited disorders, including neurofibromatosis, von Hippel–Lindau syndrome and MEN 2.

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20.46 Clinical features of phaeochromocytoma

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| • Hypertension (usually paroxysmal; often postural drop of blood pressure) | • Abdominal pain, vomiting |
| • Paroxysms of: | • Constipation |
| Pallor (occasionally flushing) | • Weight loss |
| Palpitations, sweating | • Glucose intolerance |
| Headache | |
| Anxiety (angor animi) | |

- ◆ Excessive secretion of catecholamines can be confirmed by measuring metabolites in plasma and/or urine (metanephrine and normetanephrine).
- ◆ There is a high 'false-positive' rate, as misleading metanephrine concentrations may be seen in stressed patients (during acute illness, following vigorous exercise or severe pain) and following ingestion of some drugs such as tricyclic antidepressants.
- ◆ For this reason, a repeat sample should usually be requested if elevated levels are found, although, as a rule, the higher the concentration of metanephrines, the more likely is the diagnosis of pheochromocytoma/paraganglioma.
- ◆ Serum chromogranin A is often elevated and may be a useful tumour marker in patients with non- secretory tumours and/ or metastatic disease.
- ◆ Genetic testing should be considered in individuals with other features of a genetic syndrome, with a family history of pheochromocytoma/ paraganglioma, and in those presenting under the age of 50 years.

Localisation

Pheochromocytomas are usually identified by abdominal CT or MRI

Management

- ◆ Medical therapy is required to prepare the patient for surgery, preferably for a minimum of 6 weeks to allow restoration of normal plasma volume. The most useful drug in the face of very high circulating catecholamines is the α -blocker phenoxybenzamine (10–20 mg orally 3–4 times daily) because it is a non-competitive antagonist, unlike prazosin or doxazosin.
- ◆ If α -blockade produces a marked tachycardia, then a β -blocker such as propranolol can be added.

rise in blood pressure due to unopposed α -mediated vasoconstriction.

- ◆ During surgery, sodium nitroprusside and the short-acting α -antagonist phentolamine are useful in controlling hypertensive episodes, which may result from anaesthetic induction or tumour mobilisation.
- ◆ Post-operative hypotension may occur and require volume expansion and, very occasionally, noradrenaline (norepinephrine) infusion, but is uncommon if the patient has been prepared with phenoxybenzamine.
- ◆ Metastatic tumours may behave in an aggressive or a very indolent fashion.
- ◆ Management options include debulking surgery, radionuclide therapy with ^{131}I -MIBG, chemotherapy and (chemo)embolisation of hepatic metastases; some may respond to tyrosine kinase and angiogenesis inhibitors.